DATA SCIENCE LABORATORY MANUAL

Module IV ? Statistics and Machine Learning

Topic: Unsupervised Learning

OVERVIEW & LEARNING OBJECTIVES

Clustering methods: K-Means, hierarchical clustering, and Principal Component Analysis (PCA).

KEY TOPIC CONCEPTS & METHODOLOGY

- Understanding Unsupervised Learning in modern Data Science workflows.
- Detailed breakdown: Clustering methods: K-Means, hierarchical clustering, and Principal Component Analysis (PCA).
- Mathematical formulas, code examples, and laboratory execution guidelines.
- Common pitfalls, edge cases, and best practices in practical applications.

LABORATORY PRACTICE / IMPLEMENTATION NOTES

1. Follow standard Python and Data Science coding practices.
2. Document all exploratory findings and statistical hypotheses.
3. Keep datasets and outputs organized in your repository workspace.